Experiment No. 2:Signal Operations and Transformations.

OBJECTIVE:

(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

(iii) To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics.

(i) To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

Expected Outcomes

• Correctly implement amplitude scaling.

• Perform time compression and time expansion.

• Measure the resulting amplitude and period.

• Explain the observed changes using the scaling equations.

Software / Laboratory Environment :

MATLAB

MATLAB Coding:

```
clc;
clear;
close all;

t = -1:0.001:1;

x = sin(2*pi*5*t);
yA = 2*x;
yC = sin(2*pi*5*(2*t));
```

```
yE = sin(2*pi*5*(t/2));

figure;

subplot(2,2,1);
plot(t,x,'b','LineWidth',1.5);
grid on;
title('Original Signal x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([-1 1 -2.2 2.2]);

subplot(2,2,2);
plot(t,yA,'r','LineWidth',1.5);
grid on;
title('Amplitude Scaling: 2x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([-1 1 -2.2 2.2]);

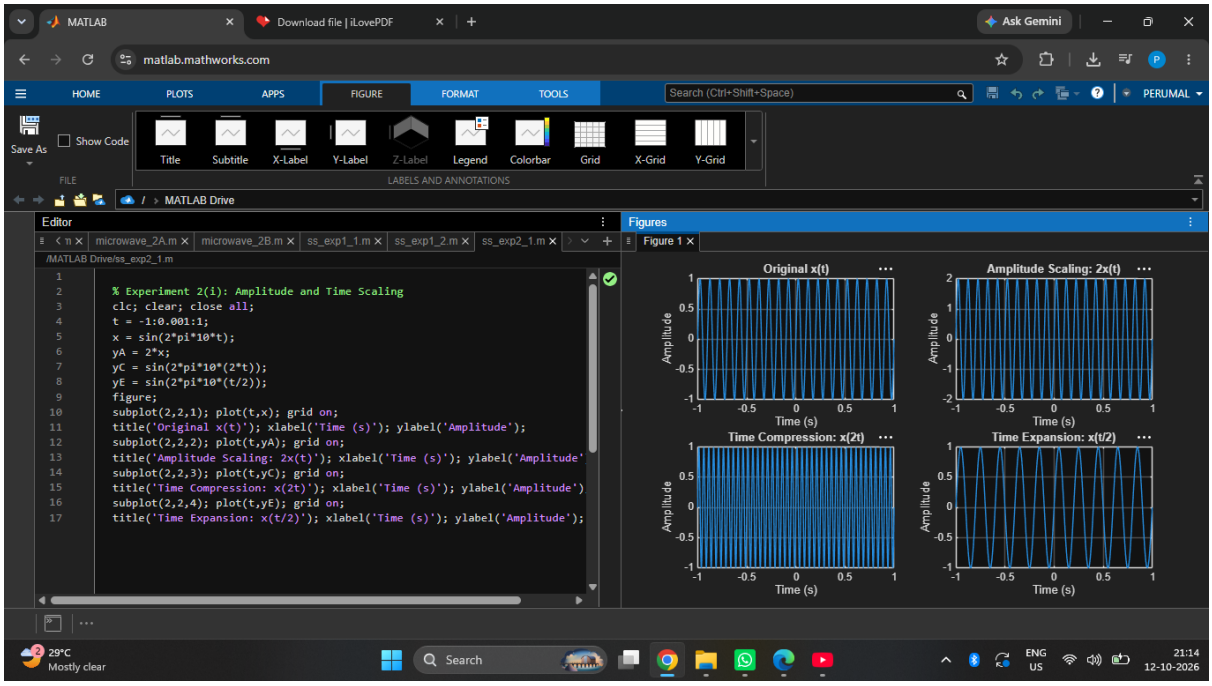
subplot(2,2,3);
plot(t,yC,'g','LineWidth',1.5);
grid on;
title('Time Compression: x(2t)');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
axis([-1 1 -2.2 2.2]);


subplot(2,2,4);
plot(t,yE,'m','LineWidth',1.5);
grid on;
title('Time Expansion: x(t/2)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([-1 1 -2.2 2.2]);


sgtitle('Experiment 2(i): Amplitude and Time Scaling');
```

OUTPUT:



RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Amplitude & Time Scaling Implementation (MATLAB)**	Generates the original and scaled versions correctly for CT and DT with correct index handling. Uses plot for CT and stem for DT, with proper time and index vectors. Original and scaled plots are placed in subplots	Generates most scaled signals correctly, with minor errors in scaling factors, index handling, axis limits or labelling.	Generates only basic scaled signals, with noticeable errors in code, time or index vectors, or plots, and needs assistance.	Unable to generate the scaled signals or produce meaningful plots.	**25**	2.5	

		with matching axes, titles, axis labels and grid.						

(ii) To perform time shifting and time reversal of continuous-time and discrete-time signals and analyze the resulting transformations.

Expected Outcomes:

• Implement delay/advance correctly.

• Generate x(−t) for time reversal.

• Identify the direction of shift from the waveform.

• Verify that time reversal preserves amplitude while reversing temporal orientation.

Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
clc;
clear;
close all;

t = -1:0.001:1;
t0 = 0.2;

x = sin(2*pi*5*t);
xShift = sin(2*pi*5*(t-t0));
xRev = sin(2*pi*5*(-t));

figure;

subplot(3,1,1);
plot(t,x,'b','LineWidth',1.5);
```

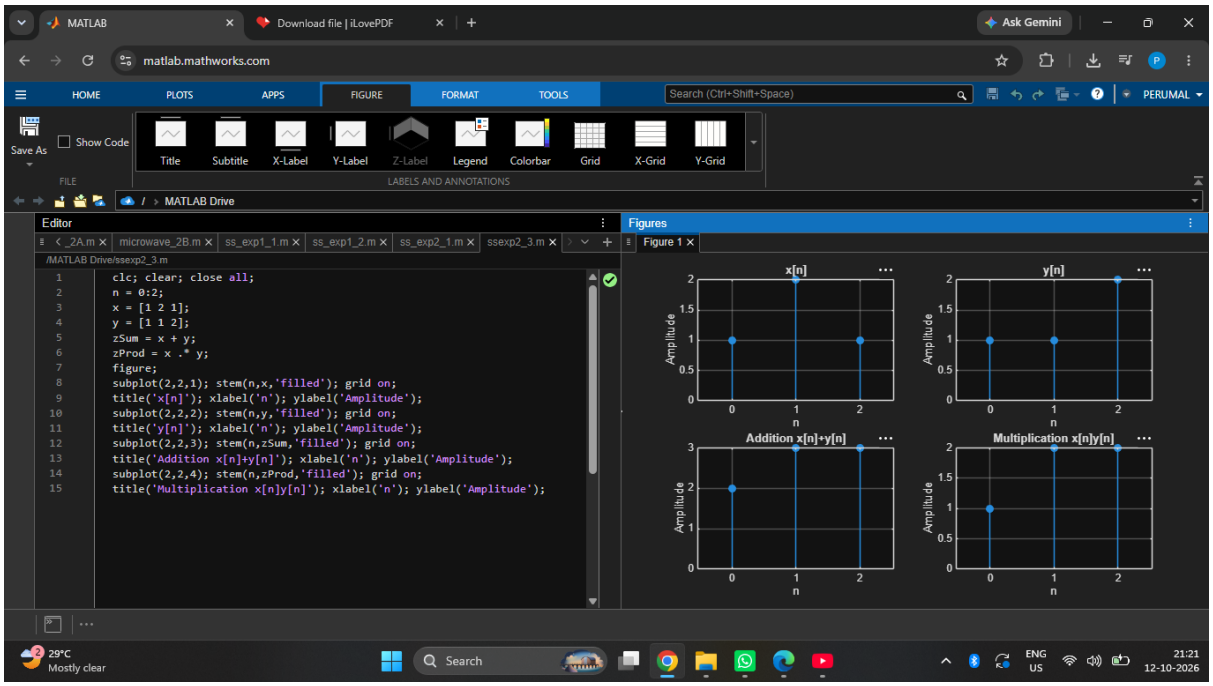
```
grid on;
title('Original Signal x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([-1 1 -1.2 1.2]);

subplot(3,1,2);
plot(t,xShift,'r','LineWidth',1.5);
grid on;
title('Time Shifting: x(t-t0), t0 = 0.2 s');
xlabel('Time (s)');
ylabel('Amplitude');
axis([-1 1 -1.2 1.2]);

subplot(3,1,3);
plot(t,xRev,'g','LineWidth',1.5);
grid on;
title('Time Reversal: x(-t)');
xlabel('Time (s)');
ylabel('Amplitude');
axis([-1 1 -1.2 1.2]);

sgtitle('Experiment 2(ii): Time Shifting and Time Reversal');
```

OUTPUT:



RESULT :

Amplitude and time scaling, sequence addition and sequence multiplication were successfully implemented.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Analysis of Scaling Effects**	Correctly explains that amplitude scaling changes amplitude and energy but not duration or period. Explains that time compression shortens duration and period (T/a) and raises frequency, that expansion does the opposite, and that a < 0 reverses time. Explains that	Explains most effects correctly, with minor errors or limited justification.	Gives basic observations on the effects. The explanation is incomplete and needs guidance.	Unable to analyze or explain the effects of scaling.	**25**	2.5	

		DT scaling discards samples and needs interpolation. Supports every point with waveform evidence.						